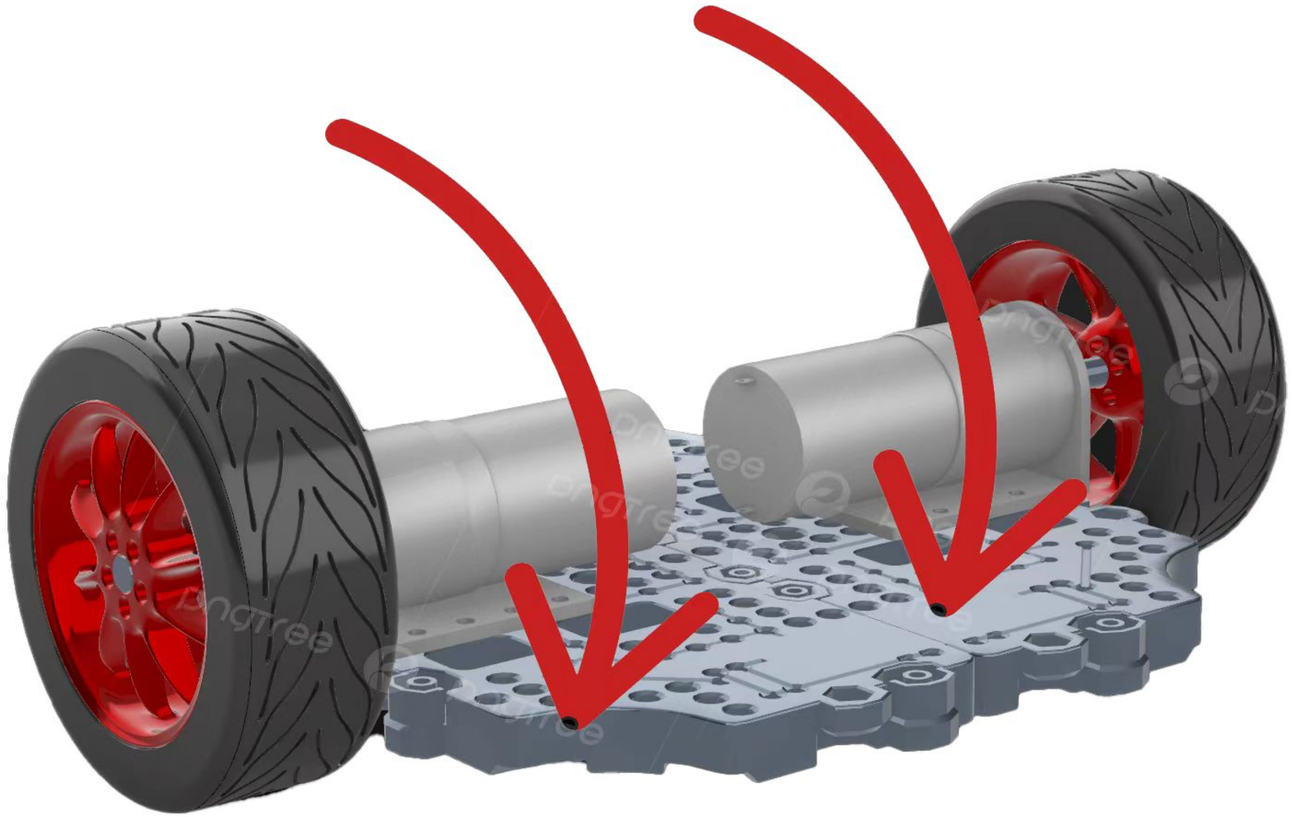
NYU

ROB-UY 3303

Robot Motion & Planning

Lecture 07A - Robot Velocity

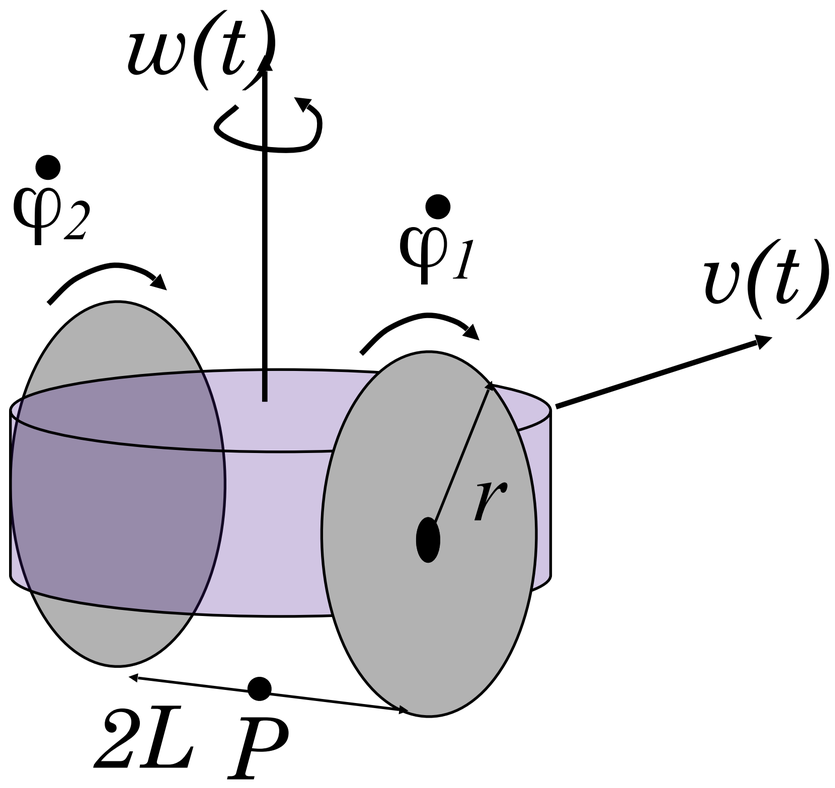


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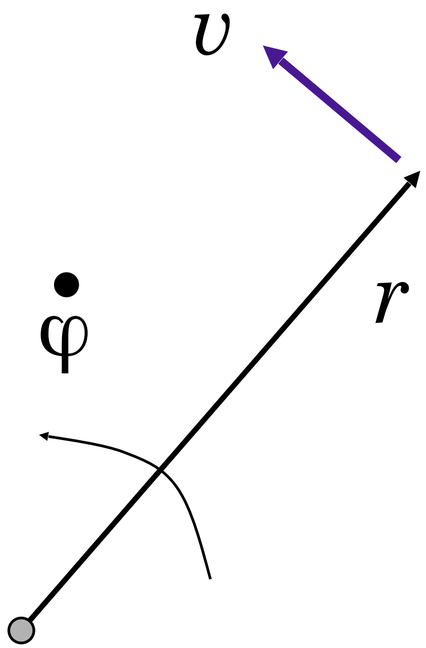
Agenda

1. Independent wheel speed
2. Robot velocity

- **Differential drive - Velocity Input**
  - The Kinematics equations should model how velocities in the global frame, are a function of wheel speed inputs – $\dot{\varphi}_1$ and $\dot{\varphi}_2$.



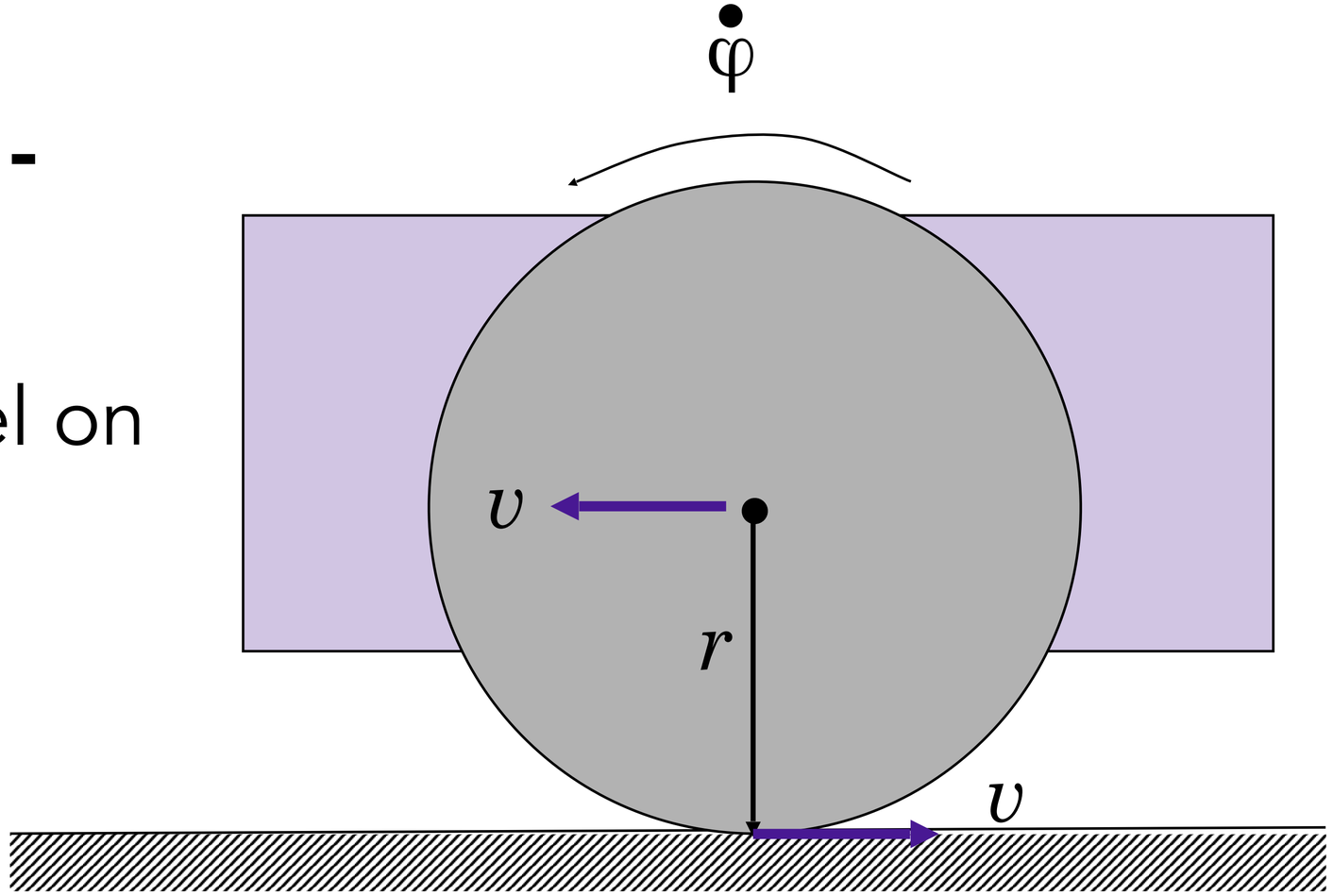
- **Differential drive - Velocity Input**
  - Before we continue, we need to understand the relation between rotational velocity and forward velocity.



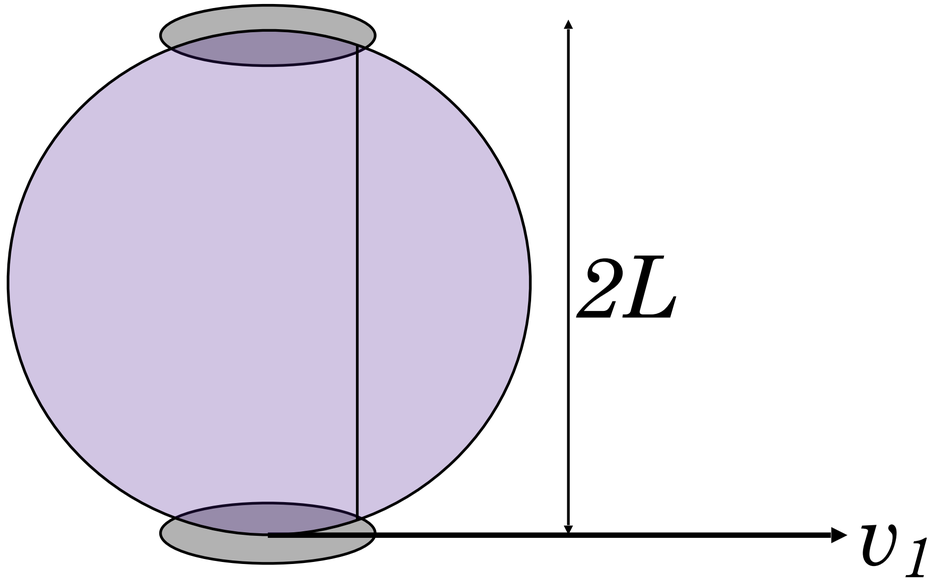
$$r\,\dot{\varphi} = v$$

- **Differential drive - Velocity Input**
  - Apply this to a wheel on the robot.

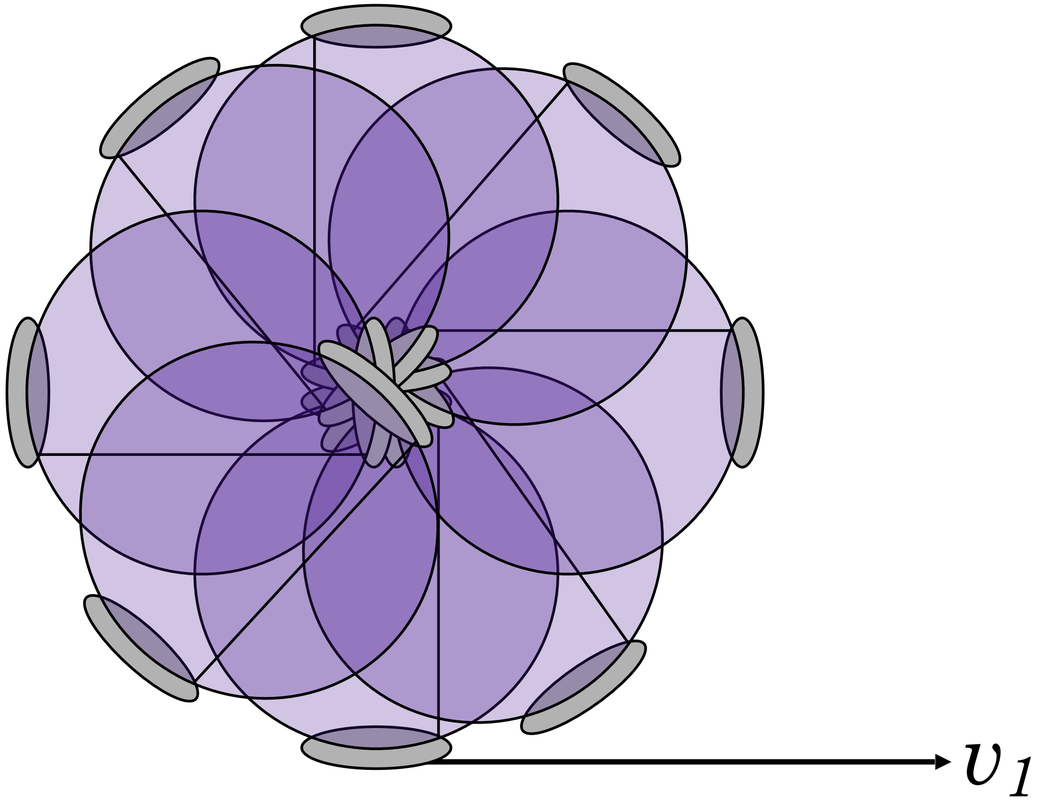
$$r\dot{\varphi} = v$$



- **Differential drive - Velocity Input**
  - Apply the same equation to a top view of the robot, assuming only wheel *1* is rotating.

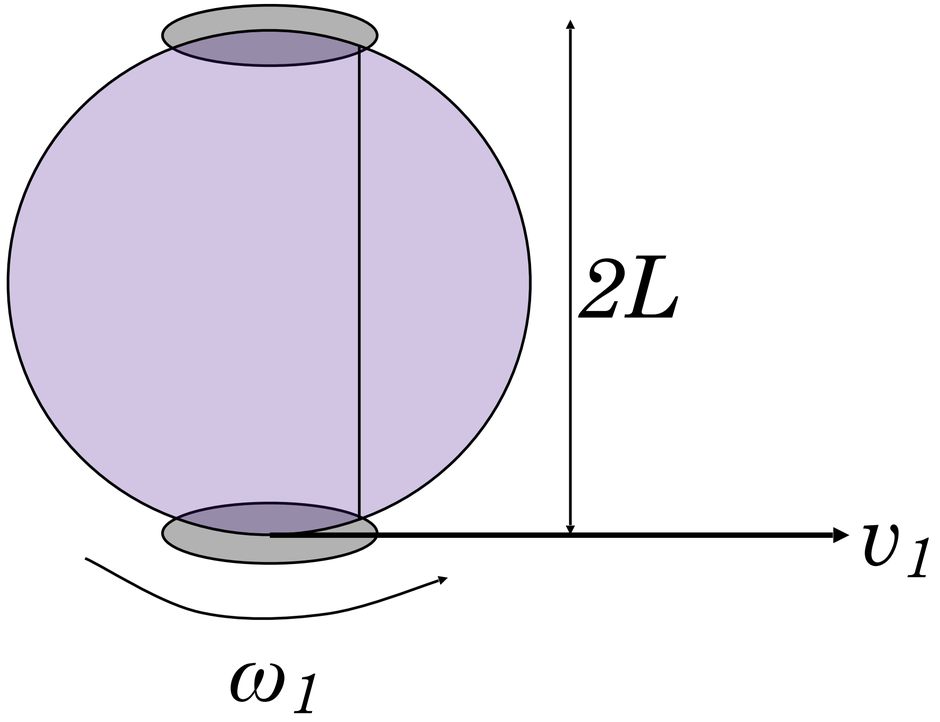


- **Differential drive - Velocity Input**
  - If the left wheel has velocity 0, and right wheel has velocity v, the robot will spin with the left wheel acting as the center of rotation.



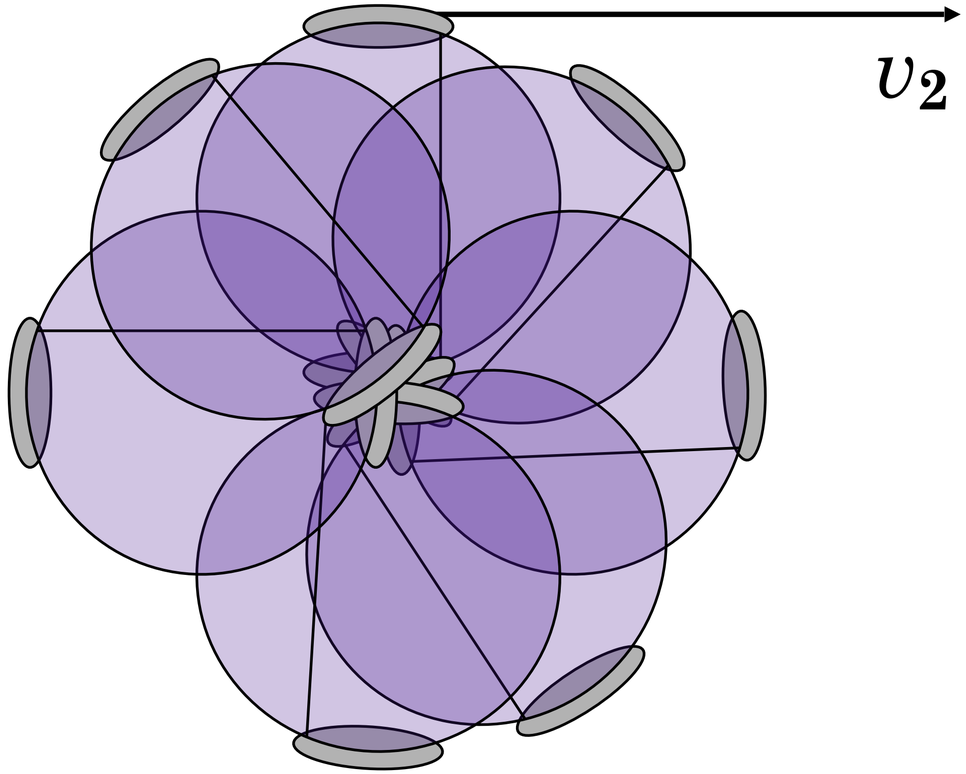
- **Differential drive - Velocity Input**
  - So the rotational velocity induced by the right wheel is:

$$\omega_1 = v_1 / 2L \quad rad/s$$



- **Differential drive - Velocity Input**
  - Similarly, the rotational velocity induced by the left wheel is:

$$\omega_2 = -v_2 / 2L \quad rad/s$$



- **Differential drive - Velocity Input**
  - Now, substitute velocities v_1 and v_2 calculated from wheel speeds into the rotational velocity equations

$$\omega_1 = \frac{r\dot{\varphi}_1}{2L}$$

$$\omega_2 = \frac{-r\dot{\varphi}_2}{2L}$$

- **Differential drive - Velocity Input**
  - Now, the rotational velocities can be calculated by summing the components of velocities from each wheel:

$$w(t) = \omega_1 + \omega_2$$

  - The forward velocity is the sum of the two components, (i.e. average of 2 velocities) again using the same equation from slide 41:

$$v(t) = L(\omega_1 - \omega_2)$$

- **Differential drive - Velocity Input**
  - Recall

$$\begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix} = \begin{bmatrix} x_{t-1} \\ y_{t-1} \\ \theta_{t-1} \end{bmatrix} + \begin{bmatrix} cos(\theta_t)\Delta t & 0 \\ sin(\theta_t)\Delta t & 0 \\ 0 & \Delta t \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}$$

- **Differential drive - Velocity Input**
  - Substituting for the velocities.

$$\begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix} = \begin{bmatrix} x_{t-1} \\ y_{t-1} \\ \theta_{t-1} \end{bmatrix} + \begin{bmatrix} cos(\theta_t)\Delta t & 0 \\ sin(\theta_t)\Delta t & 0 \\ 0 & \Delta t \end{bmatrix} \begin{bmatrix} \frac{r\dot{\phi}_1 + r\dot{\phi}_2}{2} \\ \frac{r\dot{\phi}_1 - r\dot{\phi}_2}{2L} \end{bmatrix}$$

- Differential drive - Velocity Input

$$x_{\mathbf{t}} = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix}$$

$$x_t = f(x_{t-1}, u_t)$$

- **Differential drive - Velocity Input**

$$x_t = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix}$$

$$x_t = f(x_{t-1}, u_t) = x_{t-1} + \begin{bmatrix} cos(\theta_t)\Delta t & 0 \\ sin(\theta_t)\Delta t & 0 \\ 0 & \Delta t \end{bmatrix} \begin{bmatrix} \frac{r\dot{\phi}_1 + r\dot{\phi}_2}{2} \\ \frac{r\dot{\phi}_1 - r\dot{\phi}_2}{2L} \end{bmatrix}$$